

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1-3 Semester (2025-26)
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Branch	CSE
Course Outcome	5

Case Study Topic: Zomato Delivery ETA using Quick select and Median of Medians

Word Done Summary:

In this case study, we analysed how Zomato can efficiently compute the median ETA from large delivery datasets using Quick select and Median of Medians algorithms. We studied how Quick select finds the k-th smallest element without fully sorting the array. We also examined the randomized pivot approach, worst-case complexity, and deterministic linear-time selection using the Median of Medians technique.

Implementation Details:

1. Represented delivery ETAs as an integer array.
2. Applied Quick select algorithm to find the median.
3. Performed array partitioning around pivots.
4. Analysed recursive partition selection.
5. Studied randomized pivot behaviour.
6. Examined Median of Medians deterministic selection.
7. Compared expected and worst-case complexities.

Result:

Quick select efficiently computed the median ETA without fully sorting the dataset. The algorithm demonstrated average linear-time performance, while Median of Medians provided deterministic worst-case guarantees.

Output:

```
Median ETA: 28  
  
Array after Quickselect:  
12 15 19 22 28 33 37 50 45
```

GitHub Link:

<https://github.com/ketanasindhusomisetty/DSA-Case-Studies/tree/main>

Student Signature	Faculty Signature
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